Exp 5 - Realization of IIR Filters Using Direct Form-I, Direct Form-II, Cascade and Parallel Structures Using MATLAB

Program:

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clc;
clear;
close all;

% EXPERIMENT 5
% Realization of IIR Filters Using Direct Form-I, Direct Form-II,
% Cascade and Parallel Structures Using MATLAB

%% STEP 1: INPUT PARAMETERS

Fs  = 1000;       % Sampling frequency in Hz
Fp  = 100;        % Passband edge frequency in Hz
Fst = 150;        % Stopband edge frequency in Hz
Rp  = 1;          % Passband attenuation/ripple in dB
Rs  = 40;         % Stopband attenuation in dB
N   = 128;        % Number of impulse samples

% Unit impulse input

x = zeros(1,N);
x(1) = 1;

fprintf('===== EXPERIMENT 5: INPUT PARAMETERS =====\n');
fprintf('Sampling frequency = %g Hz\n',Fs);
fprintf('Passband edge = %g Hz\n',Fp);
fprintf('Stopband edge = %g Hz\n',Fst);
fprintf('Passband ripple = %g dB\n',Rp);
fprintf('Stopband attenuation = %g dB\n',Rs);

%% STEP 2: DESIGN A BUTTERWORTH IIR LOW-PASS FILTER

Wp = Fp/(Fs/2);
Ws = Fst/(Fs/2);

[filterOrder,Wn] = buttord(Wp,Ws,Rp,Rs);
[b,a] = butter(filterOrder,Wn,'low');

fprintf('\n===== FILTER COEFFICIENTS =====\n');
fprintf('Filter order = %d\n',filterOrder);
fprintf('Normalized cutoff frequency = %.6f\n',Wn);

disp('Numerator coefficients b =');
disp(b);

disp('Denominator coefficients a =');
disp(a);

%% STEP 3: DIRECT FORM-I REALIZATION

yDF1 = zeros(1,N);

for n = 1:N
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    % Feedforward calculation
    for k = 1:length(b)
        index = n-k+1;

        if index >= 1
            yDF1(n) = yDF1(n) + b(k)*x(index);
        end
    end

    % Feedback calculation
    for k = 2:length(a)
        index = n-k+1;

        if index >= 1
            yDF1(n) = yDF1(n) - a(k)*yDF1(index);
        end
    end

end

%% STEP 4: DIRECT FORM-II REALIZATION

% Canonical Direct Form-II using an internal state sequence

order = max(length(a),length(b))-1;

aPad = [a zeros(1,order+1-length(a))];
bPad = [b zeros(1,order+1-length(b))];

v = zeros(1,N);
yDF2 = zeros(1,N);

for n = 1:N

    % Internal state
    v(n) = x(n);

    for k = 1:order
        if n-k >= 1
            v(n) = v(n) - aPad(k+1)*v(n-k);
        end
    end

    % Output calculation
    for k = 0:order
        if n-k >= 1
            yDF2(n) = yDF2(n) + bPad(k+1)*v(n-k);
        end
    end

end

%% STEP 5: CASCADE REALIZATION USING SECOND-ORDER SECTIONS

[sos,g] = tf2sos(b,a);

yCascade = g*x;
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for k = 1:size(sos,1)
    yCascade = filter(sos(k,1:3), ...
                      sos(k,4:6),yCascade);
end

fprintf('\n===== CASCADE REALIZATION =====\n');
fprintf('Number of second-order sections = %d\n',size(sos,1));
fprintf('Cascade gain = %.8g\n',g);
disp('Second-order-section coefficients:');
disp(sos);

%% STEP 6: PARALLEL REALIZATION USING PARTIAL FRACTIONS

[r,p,k] = residuez(b,a);

% Direct polynomial part, if present

if isempty(k)
    yParallel = zeros(1,N);
else
    yParallel = filter(k,1,x);
end

% Add the first-order parallel branches

for i = 1:length(r)
    branchOutput = filter(r(i),[1 -p(i)],x);
    yParallel = yParallel + branchOutput;
end

yParallel = real(yParallel);

fprintf('\n===== PARALLEL REALIZATION =====\n');
disp('Residues:');
disp(r);

disp('Poles:');
disp(p);

disp('Direct polynomial coefficients:');
disp(k);

%% STEP 7: COMPARE OUTPUTS

errorDF2 = max(abs(yDF1-yDF2));
errorCascade = max(abs(yDF1-yCascade));
errorParallel = max(abs(yDF1-yParallel));

fprintf('\n===== MAXIMUM OUTPUT ERRORS =====\n');
fprintf('Direct Form-I vs Direct Form-II = %.10g\n',errorDF2);
fprintf('Direct Form-I vs Cascade = %.10g\n',errorCascade);
fprintf('Direct Form-I vs Parallel = %.10g\n',errorParallel);

fprintf('\n===== STABILITY CHECK =====\n');
poles = roots(a);
disp('Filter poles:');
disp(poles);

if all(abs(poles) < 1)
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    disp('The designed digital filter is stable.');
else
    disp('The designed digital filter is unstable.');
end

%% STEP 8: FREQUENCY RESPONSE

[H,f] = freqz(b,a,2048,Fs);

magnitudeDB = 20*log10(abs(H)+eps);
phaseDegrees = unwrap(angle(H))*180/pi;

%% STEP 9: DISPLAY ALL GRAPHS IN ONE FIGURE WINDOW

figure('Name','Experiment 5 - IIR Realization Comparison', ...
       'NumberTitle','off');

% Graph 1: Impulse response comparison

subplot(2,2,1);

plot(0:N-1,yDF1,'b','LineWidth',1.5);
hold on;
plot(0:N-1,yDF2,'r--','LineWidth',1.2);
plot(0:N-1,yCascade,'g:','LineWidth',1.5);
plot(0:N-1,yParallel,'k-.','LineWidth',1.2);

grid on;
xlabel('Sample index n');
ylabel('Amplitude');
title('Impulse Response Comparison');

legend('Direct Form-I','Direct Form-II', ...
       'Cascade','Parallel','Location','best');

% Graph 2: Numerical error comparison

subplot(2,2,2);

semilogy(0:N-1,abs(yDF1-yDF2)+eps,'r');
hold on;
semilogy(0:N-1,abs(yDF1-yCascade)+eps,'g');
semilogy(0:N-1,abs(yDF1-yParallel)+eps,'k');

grid on;
xlabel('Sample index n');
ylabel('Absolute error');
title('Realization Error Comparison');

legend('DF-I vs DF-II','DF-I vs Cascade', ...
       'DF-I vs Parallel','Location','best');

% Graph 3: Magnitude response

subplot(2,2,3);

plot(f,magnitudeDB,'b','LineWidth',1.5);
hold on;
plot([Fp Fp],[-100 5],'r--');
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plot([Fst Fst],[-100 5],'k--');

grid on;
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
title('Butterworth Magnitude Response');
xlim([0 Fs/2]);
ylim([-100 5]);

legend('Magnitude response','Passband edge', ...
       'Stopband edge','Location','best');

% Graph 4: Phase response

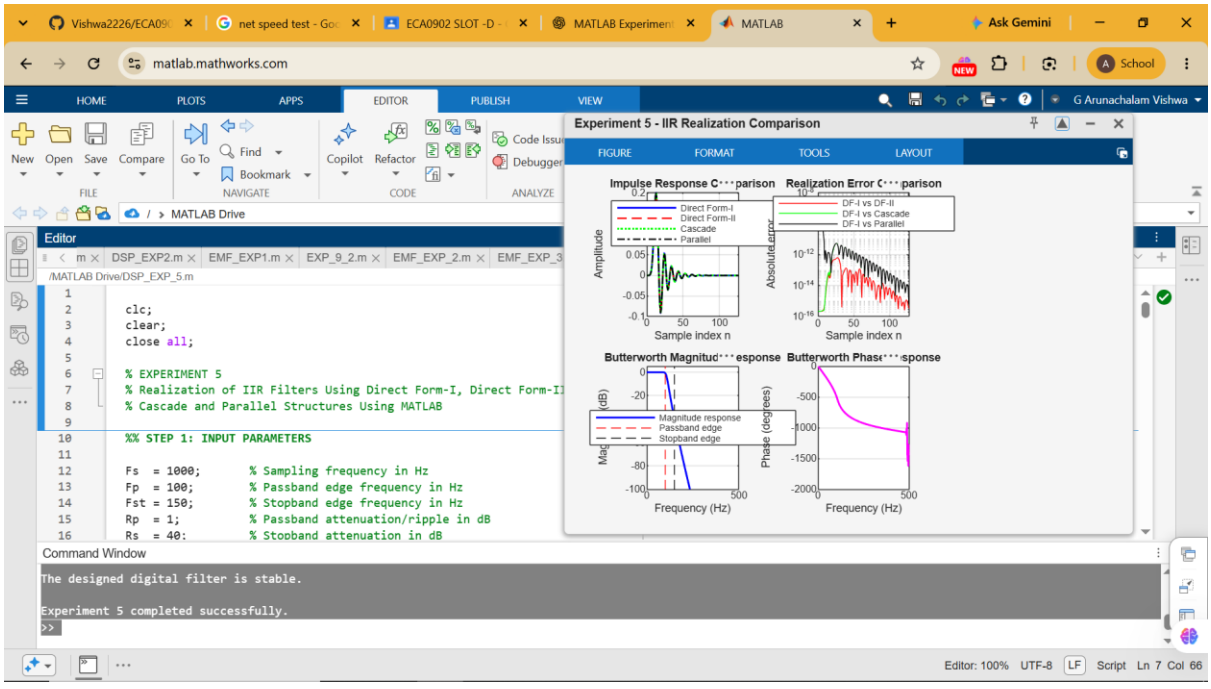
subplot(2,2,4);

plot(f,phaseDegrees,'m','LineWidth',1.5);

grid on;
xlabel('Frequency (Hz)');
ylabel('Phase (degrees)');
title('Butterworth Phase Response');
xlim([0 Fs/2]);

fprintf('\nExperiment 5 completed successfully.\n');
```

Output:



Command Window:

===== EXPERIMENT 5: INPUT PARAMETERS =====
Sampling frequency = 1000 Hz

Passband edge = 100 Hz
Stopband edge = 150 Hz
Passband ripple = 1 dB
Stopband attenuation = 40 dB

===== FILTER COEFFICIENTS =====
Filter order = 12
Normalized cutoff frequency = 0.212709
Numerator coefficients b =
1.0e-03 *

0.0002 0.0027 0.0146 0.0488 0.1098 0.1757 0.2050 0.1757 0.1098 0.0488 0.0146 0.0027 0.0002

Denominator coefficients a =
1.0000 -6.8844 22.4341 -45.5100 63.7725 -64.8501 48.9643 -27.6092 11.5218 -3.4662 0.7128 -0.0899 0.0053

===== CASCADE REALIZATION =====
Number of second-order sections = 6
Cascade gain = 2.2189438e-07
Second-order-section coefficients:
1.0000 2.1763 1.1847 1.0000 -0.9724 0.2389
1.0000 2.1201 1.1282 1.0000 -0.9983 0.2719
1.0000 2.0312 1.0389 1.0000 -1.0525 0.3409
1.0000 1.9422 0.9492 1.0000 -1.1399 0.4522
1.0000 1.8789 0.8852 1.0000 -1.2689 0.6167
1.0000 1.8514 0.8572 1.0000 -1.4524 0.8504

===== PARALLEL REALIZATION =====
Residues:
-0.1393 + 0.1132i
-0.1393 - 0.1132i
-0.6347 - 1.2462i
-0.6347 + 1.2462i
5.3136 - 1.3189i
5.3136 + 1.3189i
-0.1592 +14.0558i
-0.1592 -14.0558i
-24.7046 - 7.6372i
-24.7046 + 7.6372i
20.3242 -28.5161i
20.3242 +28.5161i

Poles:
0.7262 + 0.5683i
0.7262 - 0.5683i
0.6345 + 0.4627i

0.6345 - 0.4627i
0.5699 + 0.3569i
0.5699 - 0.3569i
0.5262 + 0.2529i
0.5262 - 0.2529i
0.4992 + 0.1508i
0.4992 - 0.1508i
0.4862 + 0.0501i
0.4862 - 0.0501i

Direct polynomial coefficients:
4.2255e-05

===== MAXIMUM OUTPUT ERRORS =====
Direct Form-I vs Direct Form-II = 6.393299518e-13
Direct Form-I vs Cascade = 5.460090713e-12
Direct Form-I vs Parallel = 9.097637388e-10

```
===== STABILITY CHECK =====
Filter poles:
   0.7262 + 0.5683i
   0.7262 - 0.5683i
   0.6345 + 0.4627i
   0.6345 - 0.4627i
   0.5699 + 0.3569i
   0.5699 - 0.3569i
   0.5262 + 0.2529i
   0.5262 - 0.2529i
   0.4992 + 0.1508i
   0.4992 - 0.1508i
   0.4862 + 0.0501i
   0.4862 - 0.0501i

The designed digital filter is stable.

Experiment 5 completed successfully.
>>
```